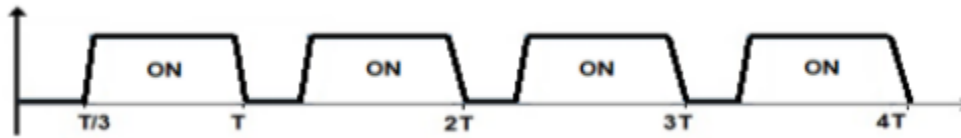


Assignment#3 (Clock cycle, timing diagram and memory banking)



1.

The diagram above shows 1 MACHINE/BUS CYCLE of an 8086 microprocessor operating at a particular frequency.

- Calculate the percentage of Duty Cycle of the waveform.
 - Calculate the frequency (in MHz) at which the 8086 is operating if each clock pulse is 'ON' for exactly 55.55 ns.
 - During a READ cycle, suppose the 8086 is operating at the frequency you got from (I) and now reading from a memory that takes 300 ns to upload data on the data bus. Explain with reasoning whether the microprocessor will insert a Wait state (T_w) or not.
2. Suppose an 8086 is operating in a way such that T_{ON} is 1/4th of the total time required for one clock pulse.

Consider $T_{ON} = 30$ ns.

Now the 8086 is going to execute the instruction MOV AX, [2315h], i.e., 16 bits of data will be read from memory.

- Calculate the frequency (in MHz) at which the 8086 is operating.
- Deduce the total time required to execute the given instruction MOV AX, [2315h].
- Explain with proper reasoning the values of pins A_0 and BHE' during the execution of the given instruction MOV AX, [2315h].
- Suppose from the start of the T_3 clock pulse, the memory took 240 ns to finalize uploading data on the data bus. Explain with reason whether the processor will exert a Wait State or not.

Assignment#4(Interrupt and I/O System Design)

1. Assume you have a different version of the 8259A Programmable Interrupt Controller (PIC) known as PICx to control the numerous hardware interrupts. The PICx can take 16 interrupt requests using the pins IR0- IR15. The Interrupt Service Register (ISR) size is X bits and the Interrupt Mask Register (IMR) is Y bits. The IR2 pin is masked by default. There are priority levels assigned to each request pin. Now suppose PICx is receiving multiple interrupt requests in the following order:

At $T = 0$ ns, an interrupt request occurred at IR12 that will take 35ns to be served.

At $T = 10\text{ns}$, IR10 and IR3 receive requests each of which takes 40ns to be served.

At $T = 25\text{ns}$, IR0 and IR2 receive requests each of which takes 50ns to be served.

- a. Deduce the values of X and Y
 - b. Explain the role of the Priority Resolver in PICx.
 - c. Explain in detail how the values of the ISR and IMR registers will change according to the above requests.
 - d. Calculate the value of T (time) when all the interrupt requests have been served.
2. Intel decided to create a new version of the 8237A DMA controller called “DMA 6”, which supports 6 channels instead of the traditional 4 channels and has 20 address pins. Additionally, the size of the port addresses has now changed to 20 bits. A special register in 8086 named “QX” keeps the value of the 20-bit port address when using Variable Addressing.
But for fixed addressing, assume the size of the P8 byte remains the same.
 - a. Deduce the maximum number of I/O devices the DMA 6 can handle.
 - b. Calculate the total size of the memory that the DMA 6 is capable of addressing.
 - c. Deduce the new range of address space for fixed addressing and variable addressing if Isolated I/O is used for I/O address mapping.
 - d. Assume DMA 6 is used in cascading mode. Illustrate with a diagram how 2 secondary DMA 6 controllers can be connected with one primary DMA 6.
 - e. Explain the complete process of how data transfer takes place between an Intel 8086 and the cascaded DMA 6 controllers.